



# The effects of critical thinking intervention on reliance behaviors, problem-solving quality, and creativity during human-Generative AI collaborative learning

Chenyu Hou<sup>a</sup>, Gaoxia Zhu<sup>b,\*</sup>, Yanzhi Liu<sup>a</sup>, Vidya Sudarshan<sup>c</sup>,  
Josephine Leng Leng Chong<sup>c</sup>, Fannie Yifan Zhang<sup>d</sup>, Michael Yong Heng Tan<sup>e</sup>, Yew  
Soon Ong<sup>c</sup>

<sup>a</sup> Nanyang Technological University, Singapore

<sup>b</sup> National Institute of Education (NIE), Nanyang Technological University, Singapore

<sup>c</sup> College of Computing & Data Science, Nanyang Technological University, Singapore

<sup>d</sup> Interdisciplinary Collaborative Core Office, Nanyang Technological University, Singapore

<sup>e</sup> College of Business, Nanyang Technological University, Singapore

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## ABSTRACT

As Generative AI becomes increasingly used in various educational contexts, understanding how students engage with these tools during collaborative problem-solving is critical. While prior research suggests that critical thinking is essential in human-AI problem-solving, few studies have examined how instructional interventions, targeting critical thinking, might shape their reliance behaviors and collaborative outcomes. This study investigates the effects of a critical thinking intervention embedded in a problem-based learning (PBL) environment where students are engaged with Generative AI. The intervention combined strategies that foster critical thinking, including authentic instruction, structured dialogue, and AI-supported peer mentoring, aiming to promote students' thoughtful engagement and improve problem-solving performance. Participants ( $N = 226$ ) were assigned to experimental (with critical thinking interventions) or comparison (without critical thinking interventions) conditions. We used pre- and post-surveys to measure participants' trust, critical thinking, and AI reliance behaviors, and group reports and chat histories to assess their problem-solving quality and creativity. Results revealed that the intervention did not produce significant improvement in self-reported critical thinking, possibly due to the short intervention duration. However, the intervention led to a marginal reduction in students' thoughtless use of Generative AI and significantly reduced the direct adoption of AI-generated content. Notably, students in the intervention condition produced more creative solutions, demonstrating higher levels of originality and idea density in their group reports. These findings suggest that *how* students use Generative AI is critical, especially when it is almost impossible to control *whether* they use it or not. The study highlights the importance of designing interventions that cultivate students' critical thinking to support creative human-AI problem-solving.

\* Corresponding author. Learning Sciences and Assessment Department, National Institute of Education (NIE), Nanyang Technological University, Singapore.

E-mail addresses: [chenyu004@e.ntu.edu.sg](mailto:chenyu004@e.ntu.edu.sg) (C. Hou), [gaoxia.zhu@nie.edu.sg](mailto:gaoxia.zhu@nie.edu.sg) (G. Zhu), [liuy0251@e.ntu.edu.sg](mailto:liuy0251@e.ntu.edu.sg) (Y. Liu), [vidya.sudarshan@ntu.edu.sg](mailto:vidya.sudarshan@ntu.edu.sg) (V. Sudarshan), [josephine.chong@ntu.edu.sg](mailto:josephine.chong@ntu.edu.sg) (J.L.L. Chong), [fannie.zhang@ntu.edu.sg](mailto:fannie.zhang@ntu.edu.sg) (F.Y. Zhang), [michael.tan@ntu.edu.sg](mailto:michael.tan@ntu.edu.sg) (M.Y.H. Tan), [asysong@ntu.edu.sg](mailto:asysong@ntu.edu.sg) (Y.S. Ong).

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## 1. Introduction

Critical thinking is essential for high-quality learning in higher education (Anggraeni et al., 2023; Benvenuti et al., 2023). In problem-based learning (PBL), where students navigate complex and ill-structured problems (O'Neil et al., 2003), critical thinking enables them to articulate reasoning, weigh alternatives, construct well-justified arguments, and justify solutions (Butterworth & Thwaites, 2013). As Generative AI is being integrated into problem-solving, tools such as ChatGPT offer new opportunities to support creativity, reasoning, and knowledge construction (Boussieux et al., 2024; Joksimovic et al., 2023). These tools can assist students in generating novel ideas, elaborating on complex issues, and organizing solutions. Rather than being viewed as a passive tool, Generative AI can function as a situational partner that co-constructs solutions with learners (Chiou & Lee, 2023).

However, the outcomes of human-AI collaboration largely depend on how students engage with AI (Pham & Sampson, 2022). Previous studies (e.g., Benvenuti et al., 2023; Zhai et al., 2024) suggest that critical thinking could improve human-Generative AI collaborative outcomes. Critically evaluating AI-generated content, including questioning assumptions, revising prompts, and integrating evidence (Hou et al., 2025; Zhai et al., 2024), might lead to more desirable outcomes such as higher-quality solutions and greater creativity. On the other hand, passive overreliance on AI-generated content might compromise collaborative outcomes (Klingbeil et al., 2024). Thus, the importance of critical thinking becomes even more pronounced in human-Generative AI collaborative problem-solving.

Building on the rationale, the present study designs and implements a critical thinking intervention embedded in a PBL context with Generative AI. Research suggests that critical thinking can be most effectively cultivated through a combination of authentic instruction, structured dialogue, and mentoring (Abrami et al., 2015), which informs the design and implementation of the critical thinking intervention embedded in a PBL framework involving Generative AI of the current study. Specifically, we examine how the intervention affects students' reliance behaviors when working with AI and how it shapes human-Generative AI collaborative outcomes, including problem-solving quality and creativity.

## 2. Literature review

### 2.1. Critical thinking and problem-solving

Educational research identifies critical thinking as a vital skill for problem-solving (Anggraeni et al., 2023). The Delphi Report defines it as purposeful judgment that supports interpretation, analysis, evaluation, and inference, grounded in evidential, conceptual, methodological, and contextual considerations (Facione, 1990).

The importance of critical thinking is further heightened with the integration of Generative AI into problem-solving processes. Gajos and Mamykina (2022) found that participants experienced learning gains when interacting only with AI explanations, in the absence of AI decisions. Conversely, in scenarios where both AI decisions and explanations were presented after people made their initial decisions and people could then revise their decisions, no learning gains were observed. These findings indicate that learning occurs when individuals make decisions by critically and independently examining available information and explanations, which highlights the crucial role of critical thinking in effective human-AI collaboration. Similarly, a study found that Generative AI can support high school students in scientific problem-solving (Tong et al., 2025). However, when students utilize Generative AI, they often use it to obtain answers rather than engaging in more collaborative exploration. This lack of critical thinking may undermine their deeper understanding of scientific concepts (Lai, 2011). Therefore, insights from both studies (i.e., Gajos & Mamykina, 2022; Lai, 2011) suggest that employing critical thinking during human-AI collaboration is essential for effective collaborative outcomes.

### 2.2. Collaborative outcomes in human-AI problem-solving

Evaluating the educational value of human-Generative AI collaboration requires close attention to its collaborative outcomes. Creativity, which reflects learners' ability to produce original ideas through human-AI problem-solving, and quality of problem-solving, which reflects their capacities to produce coherent and feasible solutions, complement each other to indicate collaborative outcomes in PBL.

Creativity, defined as the ability to generate innovative responses to complex, ill-structured problems (Jonassen, 2010; Sowden et al., 2019), is one of the desired learning outcomes of problem-solving and an important 21st-century skill (Geisinger, 2016). Emerging evidence suggests that Generative AI can aid students' creativity during problem-solving. For instance, Urban et al. (2024) found that students collaborating with ChatGPT produced more original responses and reported higher creative self-efficacy, which means that they were more confident in producing innovative ideas.

Creativity is often associated with producing responses that are divergent, unique, and original. Prior studies have evaluated creativity through human labeling of artifacts based on dimensions including fluency, flexibility, originality, and elaboration (Chang et al., 2022), as well as dimensions such as elaboration, originality, and quality (Urban et al., 2024). Recently, research has explored automatic analysis of creativity in writing through linguistic analysis and text mining. For instance, Zedelius and colleagues (2019) studied creative story writing using Coh-Metric to assess text cohesion and readability, which significantly correlate with human judgments of creativity. Likewise, semantic distance measured through Latent Semantic Analysis (LSA) can be used to evaluate novelty and essay quality by comparing texts to a training set (Latifi & Gierl, 2021).

Quality of problem-solving, on the other hand, emphasizes the cognitive strategies used to produce a well-rounded solution

including clarity, feasibility, and alignment with contextual demands (Puentes-Díaz et al., 2021). High-quality problem-solving has been linked to cognitive processes such as problem analysis, stakeholder perspective integration, solution justification, and feasibility evaluation (Kwon, Song, Sari, & Khikmatillaeva, 2019; Puentes-Díaz et al., 2021), and to collaboration processes such as negotiation of ideas and sharing of resources (Zhang et al., 2022).

While creativity and problem-solving quality are conceptually distinct, they are also interdependent. A solution must be both novel and useful to be considered truly creative (Runco & Jaeger, 2012). For example, when assessing the creativity of students' design solutions, beside originality of ideas, Chang et al. (2022) included fluency (referring to producing alternative solutions to a problem), flexibility (i.e., showing multiple possibilities), elaboration (i.e., providing details and supportive evidence of the ideas), and usefulness (i.e., the feasibility and functionality of ideas). Thus, creativity includes novelty and ideational richness, while problem-solving quality is associated with analytical rigor and contextual relevance. Together, these constructs capture both the originality and the practicality of problem-solving artifacts. Creativity and quality of problem-solving could together offer a comprehensive picture of collaborative outcomes in human-AI problem-solving.

### 2.3. Reliance on Generative AI

Reliance, in the human-AI problem-solving context, refers to the behavioral patterns of how humans interact with AI systems (Bansal et al., 2021; Bertuccelli & Cummings, 2011; Wang et al., 2008). A conceptualization of reliance focuses on the outcomes of reliance patterns: overreliance, where humans accept AI suggestions even when they are incorrect, and underreliance, where humans reject AI advice when it is correct (Passi & Vorvoreanu, 2022). However, this conceptualization ties reliance behaviors with binary outcomes (correct or incorrect), which might not be applicable in the problem-solving context where outcomes such as creativity or problem-solving quality are not binary (Fok & Weld, 2024). In ill-structured problem-solving, simplistic outcome-based definitions of reliance may obscure the cognitive processes that underpin thoughtful AI engagement (Hou et al., 2025).

To address this limitation, Hou et al. (2025) developed a self-report instrument that captures reliance behaviors independent of task outcomes. This instrument identifies four distinct patterns of Generative AI use: reflective, cautious, collaborative, and thoughtless. Reflective use involves critical evaluation and iterative refinement of AI outputs, such as revising prompts or questioning AI suggestions. Cautious use reflects active skepticism, where users detect flaws and selectively engage with AI-generated content. Thoughtless use indicates passive adoption of AI outputs without substantial scrutiny. Collaborative use captures the integration of AI contributions into group discussions and joint refinement processes. This instrument allows researchers to investigate reliance behaviors in complex problem-solving settings.

A consistent finding across studies is that higher levels of trust are associated with greater reliance on AI systems (Hoff & Bashir, 2015; Klingbeil et al., 2024). Trust is commonly defined as the belief that an agent will assist in achieving personal goals in situations marked by uncertainty and vulnerability (Lee & See, 2004). Excessive trust can lead users to accept AI recommendations even when they are contradicted by other evidence (Klingbeil et al., 2024). Conversely, insufficient trust may cause users to dismiss accurate AI suggestions, resulting in under-reliance and diminished performance in human-AI collaboration (Hoff & Bashir, 2015). Thus, trust is an important predictor of reliance in human-AI collaboration, but it is not its determinant. In addition, people tend to engage in a cognitive evaluation process prior to accepting or rejecting AI suggestions. Vasconcelos et al. (2023) demonstrated that individuals strategically change their reliance on AI based on a cost-benefit evaluation of cognitive effort and task demands. By varying the complexity of AI explanations and reward structures, they showed that participants were more likely to engage critically with AI when the cognitive costs were manageable and the perceived utility was high.

These findings highlight that reliance is not fixed but shaped by cognitive and individual factors (Hou et al., 2025), suggesting that learners can be guided to engage with AI more thoughtfully. This underscores the need for targeted interventions that cultivate critical thinking, enabling students to manage their reliance on AI in ways that enhance problem-solving and learning outcomes.

### 2.4. Interventions to foster critical thinking in problem-solving

A review (Abrami et al., 2015) demonstrated that combining different strategies showed the most substantial and sustained improvements in students' critical thinking. Specifically, the three effective instructional strategies identified in the review are: authentic instruction, dialogue, and mentoring (Abrami et al., 2015).

Authentic instruction, which situates learning within real-world, ill-structured problems, has been identified as a particularly effective strategy for promoting critical thinking (Abrami et al., 2015). Meta-analysis also demonstrated that collaborative problem-solving tasks involving authentic, poorly structured real-world problems yielded a substantial overall improvement in students' critical thinking (Xu et al., 2023). For example, Yang and Chou (2008) examined college students collaborating on web-based bulletin boards to debate and reflect on controversial real-world issues, such as environmental protection and social justice, leading to measurable improvements in their critical thinking skills. These findings imply the importance of designing authentic, sustained problem-solving experiences to cultivate students' critical thinking. When Generative AI is involved, without real-world problems that demand critical analysis and evaluation, students may default to passively accepting AI-generated outputs, leading to thoughtless reliance behaviors.

In addition, dialogue has been identified as a critical mechanism for fostering students' critical thinking skills. Abrami et al. (2015) emphasized that learning through dialogue, including teacher-led questioning, small group discussions, or Socratic exchanges, encourages students to clarify ideas, challenge assumptions, and reason through multiple perspectives, leading to significant gains in critical thinking. Role-play activities, by design, create structured opportunities for this type of dialogue. As Latif et al. (2018)

demonstrated, role-play during PBL sessions prompts students to engage in authentic exchanges by assuming roles, negotiating meaning, debating positions, and collaboratively constructing solutions to complex scenarios. Thus, role-play can be viewed as an effective intervention strategy for promoting critical thinking through fostering structured dialogues.

Last but not least, mentoring that offers personalized feedback and scaffolds learners' development can foster critical thinking (Abrami et al., 2015). Effective mentoring helps students internalize metacognitive strategies by providing timely error correction, modeling reflective practice, and guiding students' reasoning through complex problems (Mooney Simmie & Moles, 2011). Recent technological advancements have opened new possibilities for scalable mentoring through Generative AI chatbots. For example, Favero et al. (2024) introduced a Socratic chatbot fine-tuned on open-source language models to foster critical thinking. This Socratic chatbot engages learners through structured, thought-provoking questions designed to stimulate reflection, challenge assumptions, and encourage exploration of multiple perspectives. Thus, Generative AI chatbots designed to implement Socratic questioning principles may serve as an innovative form of mentoring, offering personalized support that mirrors the essential elements of human mentoring.

## 2.5. The current study

In summary, there is growing attention to critical thinking when integrating Generative AI in educational settings to ensure productive human-Generative AI collaboration. However, few empirical studies have tested whether a critical thinking intervention can intentionally shape students' reliance behaviors with AI in authentic classroom problem solving. Addressing this gap, we designed a critical thinking intervention in the PBL context with three strategies informed by previous research (e.g., Abrami et al., 2015; Liu & Pásztor, 2022), including authentic instruction, dialogues, and mentoring. The intervention includes a clear focus on critical thinking and its relationship to Generative AI use (authentic instruction), a role-playing element for interdisciplinary communication (dialogue), and a GPT-based peer tutor to offer constructive feedback (mentoring). In contrast, the comparison condition allows for free use of Generative AI for problem-solving.

We investigated the intervention effect across two levels. First, on the individual level, we examined whether the designed intervention could improve students' perceived critical thinking, and how the intervention influences students' reliance patterns on AI. Considering that the participants in the current study had diverse disciplinary backgrounds, and previous studies have shown that trust is an important predictor of reliance behaviors, we included disciplinary backgrounds and trust in Generative AI as control variables. Second, on the group level, we investigated how the intervention influences the quality and creativity of students' collaborative problem-solving outcomes. By systematically exploring these relationships, this study contributes to a deeper understanding of how to foster thoughtful engagement with AI technologies. Specifically, the research questions are:

- RQ1 Does the intervention improve students' perceived critical thinking during human-AI problem-solving, considering individual differences in trust and disciplinary background?
- RQ2 How does the intervention impact individual students' reliance behaviors (i.e., reflective use, collaborative use, thoughtless use, and cautious use) on Generative AI, considering the effect of trust and disciplinary background?
- RQ3 On the group level, how do human-AI collaborative outcomes, including problem-solving qualities, adoption of Generative AI's outputs, and creativity, correlate with each other?

**Table 1**  
The demographic information of the participants.

|                                | Comparison condition | Experiment condition |
|--------------------------------|----------------------|----------------------|
| <b>Total</b>                   | 112                  | 114                  |
| <b>Gender</b>                  |                      |                      |
| Female                         | 60                   | 53                   |
| Male                           | 49                   | 59                   |
| I prefer not to say            | 3                    | 2                    |
| <b>Age</b>                     |                      |                      |
| Less and Equal to 18           | 10                   | 11                   |
| 19 to 22                       | 97                   | 88                   |
| 23 and above                   | 5                    | 15                   |
| <b>Academic Year</b>           |                      |                      |
| Year 1                         | 98                   | 102                  |
| Year 2                         | 14                   | 12                   |
| <b>Ethnicity</b>               |                      |                      |
| Chinese                        | 88                   | 83                   |
| Indian                         | 12                   | 6                    |
| Malay                          | 6                    | 12                   |
| Others                         | 4                    | 11                   |
| Prefer not to say              | 2                    | 2                    |
| <b>Disciplinary Background</b> |                      |                      |
| STEM                           | 84                   | 86                   |
| Non-STEM                       | 25                   | 28                   |
| Prefer not to say              | 3                    | 0                    |

RQ4 On the group level, how do human-AI collaborative outcomes, in terms of quality of problem-solving, adoption of Generative AI's outputs, and creativity, differ between conditions?

### 3. Methods and materials

#### 3.1. Participants

The participants of this study were 226 undergraduate students. They were first- and second-year students (enrolled in the university in 2023 or 2024) enrolled in a digital literacy course in the August semester of 2024 at an Asian university. The participants came from 14 different classes, with around 20 to 50 students in each class. In each class, 5 to 6 students from different schools were purposefully assigned to a group that would work on various problem-solving activities. We chose one problem-solving activity on drafting a policy proposal that allowed students to use Generative AI extensively (Please see [Appendix A](#) for the complete problem-solving activity used in this study). We assigned the 14 classes to the experiment and comparison conditions, and the students in the same class were in the same condition. In the experiment condition, students received support from three strategies to promote critical thinking (Please see Intervention Design for Critical Thinking for details). The comparison group completed the same task but used Generative AI without these supports. The demographic information of the 226 individual participants in the comparison and intervention conditions is presented in [Table 1](#). We have 112 participants in the comparison condition and 114 participants in the intervention condition. The distributions of age, gender, academic year, ethnicity, and disciplinary backgrounds are roughly balanced across the two conditions. For disciplinary backgrounds, STEM disciplines include engineering, computer science, physics, mathematics, and medical programs. Non-STEM includes social sciences, humanities, and business programs. Considering that STEM students usually have higher AI literacy than non-STEM students ([Kong et al., 2021](#)), we included disciplinary background as a control variable for the individual-level analysis.

#### 3.2. Critical thinking intervention design

The study involves a critical thinking intervention to promote human-AI collaborative outcomes. The intervention was implemented in 14 classes in a 2-hour in-class session. The study followed a between-subjects experimental design, with classes assigned to either the experiment condition or the comparison condition. The goal of the intervention was to improve students' critical thinking and promote more thoughtful reliance on Generative AI during collaborative problem-solving ([Klingbeil et al., 2024](#)). Informed by [Abrami et al.'s](#) meta-analysis ([2015](#)), the intervention incorporates three core strategies: authentic instruction, AI-mediated peer tutoring, and role-play. In contrast, the comparison condition followed the same problem-solving structure but allowed students to use Generative AI freely without any additional scaffolding.

First, for authentic instruction, students were guided to identify potential biases in AI-generated content *before* beginning the problem-solving task. This critical thinking exercise explicitly highlighted the importance of critical thinking when using Generative AI. Students were tasked with creating prompts that may expose AI biases and analyzing the Generative AI outputs to identify the biases. In addition, as outlined in the instructional design, the activity centered on real-life, complex scenarios relevant to students' experiences. This approach is grounded in prior empirical studies showing that authentic problems are effective in fostering critical thinking ([Abrami et al., 2015](#)).

Second, we designed a GPT-based peer tutor to support problem solving (see [Fig. 1](#)). Mentoring, such as teachers' feedback and tutors' facilitation, fosters critical thinking in problem-solving ([Abrami et al., 2015](#)). Since instructors may find it challenging to track all groups' progress and provide timely and personalized mentoring during class, we developed the GPT-based peer tutor to help students with this problem-solving task. This customized chatbot, using the GPT model, interacted with students by posing reflective questions, identifying logical gaps, and suggesting counterarguments ([Open AI, 2025](#)). As presented in [Fig. 1](#), specifically, the chatbot was customized to 1) identify students' problem-solving stages to provide adaptive feedback (e.g., "I understand that you are currently working on Step 3 – Present and evaluate the solution, because you are asking for feedback on what you've proposed", see [Fig. 1a](#)); 2) promote critical thinking by asking Socratic questions (e.g., "What interests of which stakeholders might be compromised in your solution?", "What are some possible causes or contributing factors to the issue?", see [Fig. 1b](#)); 3) monitor the problem-solving progress (e.g., "You've outlined clear steps for fairness ... To strengthen your evaluation, could there be any unintended consequences?", see [Fig. 1c](#)). This method draws on the effectiveness of AI-mediated Socratic tutoring in fostering critical thinking among students ([Fakour & Imani, 2025](#)).

Third, we implemented a role-play component to foster dialogue among students in the problem-solving activity. Group members were assigned as different stakeholders, including administrators, instructors, and students, and then they were required to adopt their assigned stakeholder perspectives to evaluate ethical and pedagogical issues in real AI misuse cases. Role-play is recognized as a productive strategy to promote critical thinking, engaging students cognitively and affectively to resolve issues ([Rashid & Qaisar, 2017](#)).

#### 3.3. Procedures

A complete flow of the study protocol is presented in [Fig. 2](#), including intervention designs, procedures, and data collection. In the week prior to the intervention, instructors attended a briefing session to familiarize themselves with the two versions of lesson plans and students' handouts for the experiment and comparison conditions. The handout detailed the problem-solving requirements, case



ICC002 - Peer Tutor 40 ▾

Give me some constructive feedback on my solution

Hey, thank you for your question! I understand that you are currently working on **Step 3 – Present and evaluate the solution**, because you're asking for feedback on what you've proposed. I'd love to help!

To start:

**Can you briefly summarize your solution for me?**

Also, **what core issue are you aiming to address with this solution?**

Once I have that, I can offer specific and constructive feedback! 🥰

📄 👍 🗨️ ⋮

Our solution is to educate students on AI tools. By teaching them about their limitations, students will have a greater understanding of AI and its weaknesses. This will allow them to use these tools more responsibly and effectively.

Thanks for sharing your solution! It sounds like you're focusing on **raising awareness and responsible**

(a) Identify students' problem-solving stage

ICC002 - Peer Tutor 5 ▾ 🔗 Share ⋮

can you give me a solution?

Thanks for your question! Since you're asking for a solution, it sounds like you're now moving into **Step 2: Generate the solution**—great progress! 🥰

But before we create a solution, let's pause and reflect:

1. What **core issue** did you identify in the case during Step 1?
2. What are some possible **causes or contributing factors** to that issue?

Once we're clear on those, we can brainstorm a stronger, more targeted solution together!

📄 👍 🗨️ ⋮

the fairness issue among group members

Thanks for sharing that! So you've identified **fairness among group members** as the core issue—nice job spotting a key challenge many students can relate to. 👍👍👍

(b) Promote critical thinking

ICC002 - Peer Tutor 5 ▾ 🔗 Share ⋮

Fair division of tasks: Group members agree early on who will do what, and cross-check each other's work to provide peer solution

Contribution tracking: Use shared documents or tools (like Google Docs or Trello) to make work visible.

Dual responsibility: Group ensures communication is open; instructor checks in regularly or provides a reflection form to verify individual contributions./ instructors present fair grading and clear instructions about using Gen AI

This looks very solid! 🥰 You've outlined clear steps for fairness:

- **Fair division & peer cross-checking** → builds accountability.
- **Contribution tracking** → ensures transparency.
- **Dual responsibility with instructor guidance** → supports fairness and sets boundaries with Gen AI.

To strengthen your evaluation:

1. What do you think is the **biggest strength** of this solution?
2. Could there be any **unintended downsides** (like extra workload from cross-checking or over-reliance on the instructor)?

📄 👍 🗨️ ⋮ ⬇️

(c) Monitor the problem-solving progress

Fig. 1. The design and interface of the GPT-based peer tutor

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scenarios, and scaffolding questions. During the briefing session, the researchers also explained the intervention designs and the rationale.

In the week prior to the intervention, the research team conducted a 15-minute in-class recruitment session across 14 classes. Hardcopy consent forms were distributed, and students were given the opportunity to ask questions. Participants were informed to make individual choices on whether they wished to participate in the study, and their group-level data would not be included in the analysis if the research team did not obtain consent from every group member. Those who consented to participate then completed a pre-survey online, which included demographic questions and a trust in Generative AI scale.

The intervention was delivered in one session of the digital literacy course. The instructors first reviewed concepts related to AI, such as machine learning, deep learning, and potential AI applications. After the review, all students engaged in the same problem-solving activity (Please see [Appendix A](#) for details). The problem-solving task involves drafting a policy proposal for real-life cases of AI misuse in academic contexts. For both conditions, the activity consisted of three stages: problem identification, information exploration, and solution presentation. In the first stage, students worked in teams to select a case from two scenarios and identify the core issue. They analyzed the impact on learning experiences and course outcomes and articulated stakeholder concerns from instructors, peers, and administrators. In the second stage, students proposed strategies to address the core issue, ensuring alignment with stakeholders' needs. They assessed the strengths and weaknesses of their solutions and reflected on potential conflicts of interest. In the final stage, students presented and evaluated their solutions, emphasizing reflection and improvement. They assessed if their solutions addressed the initial problem, considered consequences, and discussed how to measure effectiveness with success indicators. As explained in the Critical Thinking Intervention Design, the experiment groups received the three instructional scaffolds to help them engage with Generative AI critically, while the comparison group could use Generative AI freely. Following the session, all participants completed a reflection survey that captured their critical thinking and reliance behaviors. They also submitted a group report summarizing their problem-solving solutions, their chat histories with Generative AI, and reflections.

## 4. Data collection

### 4.1. Individual level data

**Survey.** We collected 226 survey responses. Before the in-class activity, students completed a survey on their demographic

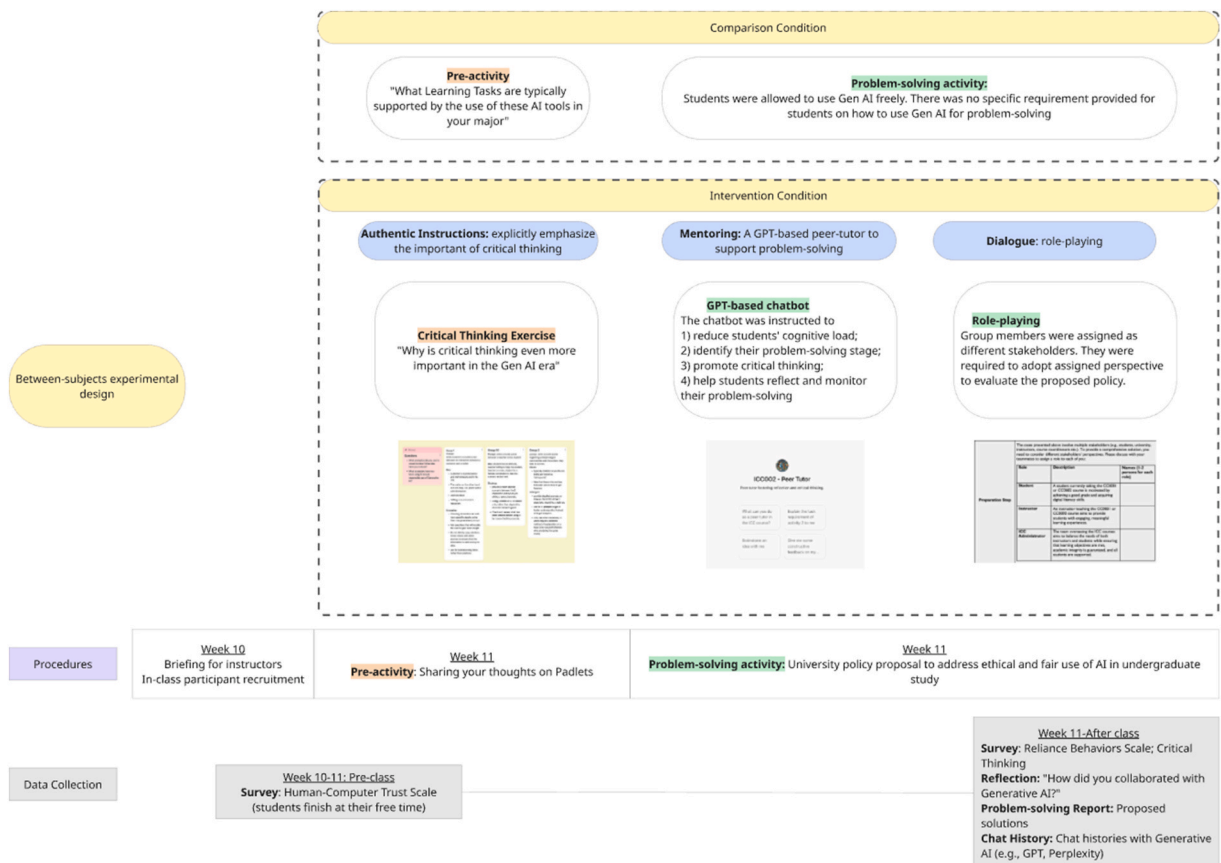


Fig. 2. An overview of the study protocol.

information (i.e., age, gender, school year, major and ethnicity) and trust level towards Generative AI. We adopted the Human-Computer Trust Scale (Gulati et al., 2019), which evaluates the degree of trust users place in automation, to assess students' trust in Generative AI for real-world problems and complex tasks. The scale comprises 12 items measured on a 5-point Likert scale (1 = strongly disagree; 5 = strongly agree). Example items include "I feel I must be cautious when using Generative AI in problem-solving" and "I think Generative AI is competent and effective in helping me solve real-world problems." The scale has been validated with good reliability (Cronbach's alpha = .84).

After the in-class activity, participants were instructed to complete the Reliance Behaviors Scale and Critical Thinking Skills. The Reliance Behaviors Scale was created and validated by the research team (Hou et al., 2025) in a problem-solving environment with similar participants. This scale assesses how frequently different usage behaviors of Generative AI are adopted, utilizing a 5-point Likert Scale (1 = Almost never, 5 = Almost always). It comprises 16 questions divided equally into four subcategories: reflective use (e.g., "I read Generative AI's output critically when the output was generated"), cautious use (e.g., "I found Generative AI's output is not perfect"), thoughtless use (e.g., "I copied the Generative AI's output as part of our solution"), and collaborative use (e.g., "My peer (s) helped me modify Generative AI's output"). According to the authors' earlier study validating this scale (2025), it demonstrates good reliability and validity (Cronbach's alpha = .84).

The Critical Thinking Skills scale was adopted from the Motivated Strategies for Learning Questionnaire (MSLQ; Pintrich et al., 1993). Individual MSLQ subscales have been utilized to assess various aspects of the learning experience (Credé & Phillips, 2011). The Critical Thinking Skills subscale specifically evaluates how well students use their knowledge in problem-solving or conduct critical assessments to develop practical and comprehensive solutions for given tasks. It employs a 7-point Likert Scale (1 = not true of me at all; 7 = very true of me) and comprises 5 items, including statements like "I often find myself questioning things I hear or read in this activity to decide if I find them convincing" and "I try to play around with ideas of my own related to what I am learning in this course." According to the original research (Pintrich et al., 1993), this scale shows good reliability and validity (Cronbach's alpha = .80).

#### 4.2. Group-level data

For group-level data, we collected participants' group reports, which included their proposed solutions and reflections on their use of Generative AI during problem-solving, as well as chat histories with Generative AI. The data at the group level requires all the group members' consent to be included in the study. We collected 38 problem-solving reports and reflections, and 29 of them also provided their chat histories with Generative AI (e.g., ChatGPT, Bard, Perplexity).

**Problem-solving Reports.** The problem-solving reports include groups' proposed solutions to address the cases of using Generative AI in collaborative projects. The cases are about issues related to Generative AI during teaching and learning in undergraduate studies. In addition, the problem-solving reports include their responses to the scaffolding questions, which were designed to facilitate students in developing and evaluating the solutions. The scaffolding questions include "What is the core issue in the chosen case?", "How do the actions in your solutions align with the needs of the stakeholders?", and "What are the potential positive and negative outcomes of implementing this solution?"

**Reflection.** In the group reports, the students were also required to reflect on their AI use. The questions include: "For what purposes have you used Generative AI in this problem-solving task?", "What aspects of Generative AI outputs did you find helpful?", "What aspects of Generative AI's output did you find not helpful?", and "How did you improve Generative AI's output?"

**Chat histories with Generative AI.** At the end of the group reports, students were required to submit their chat histories with AI

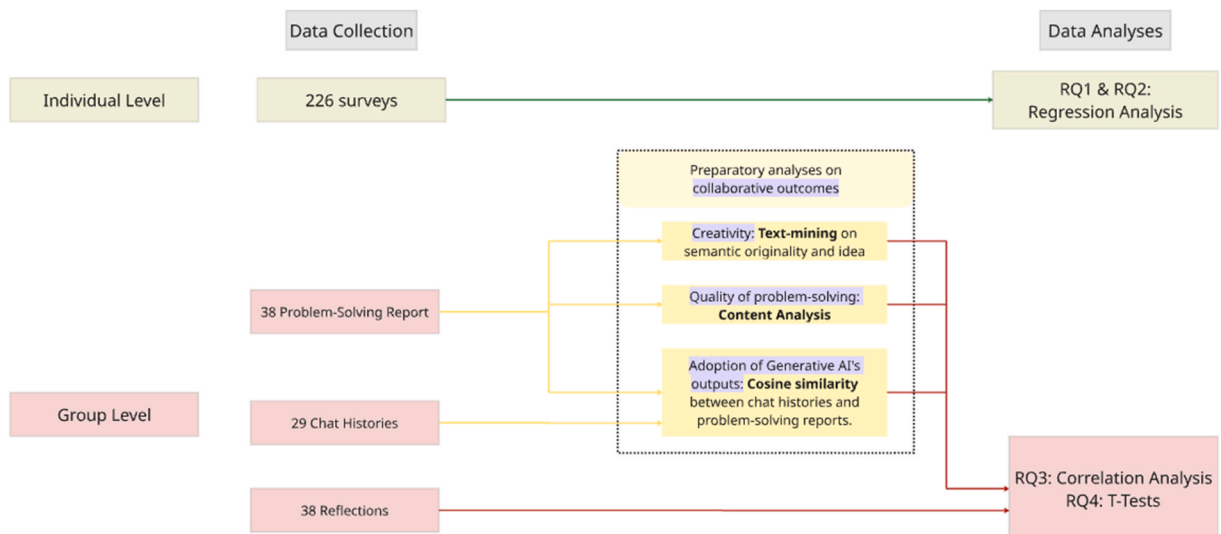


Fig. 3. An overview of data analyses.



tools using links or screenshots. Since some students did not submit the link or some links were inaccessible due to students deleting them, we included 28 chat histories in the analysis.

#### 4.3. Data analyses

To answer the research questions, we performed a series of data analyses (Please see Fig. 3 for an overview of the analyses). We performed analysis at two levels: individual-level analysis using individual surveys and group-level analysis, based on group reports that included chat histories, problem-solving reports, and reflections. We separated these two levels of analysis because of the conceptual distinction between individual-level constructs (e.g., trust, reliance behaviors) and group-level outcomes (e.g., creativity, solution quality). Aggregating individual data to the group level or disaggregating group-level data to individuals might compromise validity if within-group variability is not properly accounted for (Bliese, 2000).

First, to investigate whether the intervention has changed students' perceived critical thinking, we conducted a regression analysis controlling for the effect of trust and disciplinary backgrounds (RQ1). Second, to examine the effect of interventions on individual reliance behaviors with trust and disciplinary backgrounds as control variables, we conducted a series of regression analyses (RQ2). Third, for group-level analyses that investigate human-AI collaborative outcomes (RQ3 & RQ4), we conducted several preparatory analyses. For the quality of problem-solving, we conducted a content analysis. Then, to assess the creativity of ideas in students' problem-solving reports, we conducted two text-mining techniques (i.e., semantic similarity and idea density metrics; Bower et al., 2023) and human rating of novelty. We also computed the similarity between students' reports and their chat histories with Generative AI as an indicator of their adoption of Generative AI's output. After these preparatory analyses, we conducted correlation analyses among these indicators of collaborative outcomes (RQ3). Further, to compare collaborative performances between the conditions, we conducted a series of t-tests for each collaboration indicator from the adoption of Generative AI's output, the quality of problem-solving reports, and from creativity (RQ4). The reflection was used to supplement the quantitative analyses on the comparison of group-level collaborative outcomes.

#### 4.4. Quality of problem-solving reports

We first conducted a content analysis of students' problem-solving reports to evaluate their problem-solving performance. As presented in Appendix B, the coding scheme was developed in alignment with the learning objectives of the interdisciplinary undergraduate course designed to foster communication across disciplines, and with the core goals of PBL, which emphasize critical analysis, higher-order thinking, and evaluative reasoning (Hmelo-Silver, 2004; Vidgor & Krupnik-Gottlieb, 2015).

Effective problem-solving begins with a clear and comprehensive understanding of the problem. In the problem description stage, *Problem Definition* assesses students' ability to identify and articulate the core issue in the case study. According to Jonassen (2000), problems in everyday and professional contexts are inherently situated within complex environments, which require students to distinguish relevant from irrelevant information and to construct a meaningful problem space. Additionally, *Impact Analysis* assesses students' ability to analyze how the identified problem impacts learning experiences and course outcomes. Assessing the broader impact of a problem is essential for prioritizing actions and anticipating consequences (Hmelo-Silver, 2004). *Perspective-Taking* measures students' recognition and integration of different stakeholders' concerns. In PBL, it is necessary to gather knowledge from different disciplines to enhance the depth and quality of solution, during which students also develop skills such as communication and negotiation (Hmelo-Silver, 2004).

In the second stage, problem exploration, students start to draft solutions to address the problem. After students identify different perspectives in the first stage, it is important to align proposed solutions with stakeholders' needs to ensure relevance and feasibility. *Solution Alignment* assesses the extent to which students' solutions address the concerns of all relevant parties. *Evaluation* examines students' ability to assess their solutions' strengths, weaknesses, and potential improvements. Studies have highlighted that reflective evaluation enhances problem-solving skills by promoting deeper understanding (Noroozi et al., 2016). *Conflict Analysis* evaluates students' recognition of potential disagreements among the stakeholders they represent. Students need to be aware that their solutions would require compromises from stakeholders, and they need to effectively identify these conflicts, which is an important aspect of collaboration skills (Chinn & Malhotra, 2002). By navigating conflicts, students can enhance the robustness of their solutions.

In the third stage, solution presentation, the students need to present the solutions and evaluate the feasibility. *Solution Justification* assesses students' ability to support their proposed solutions with logical reasoning and evidence. Justifying solutions is linked to critical thinking and the ability to communicate complex ideas effectively (Vidgor and Krupnik-Gottlieb, 2015). *Refinement* measures students' ability to revise their solutions in response to identified weaknesses in the second stage. It is expected that students would refine their solution, which could enhance their learning and problem-solving capabilities (Sato, 2022). *Feasibility* evaluates students' consideration of implementation challenges and resource constraints. Assessing the practicality of solutions ensures their viability in a real-world context. *Measurable Success* examines students' ability to establish clear criteria for assessing solution effectiveness and impact.

Using the consolidated coding scheme, two raters independently evaluated the 38 reports collected from student groups. The inter-rater reliabilities between the two-raters are as follows: .63 for problem definition, .62 for impact analysis, .88 for perspective taking, .51 for solution alignment, .96 for evaluation, .80 for conflict analysis, .68 for solution justification, .48 for refinement, .80 for feasibility, and .76 for measurable success. The inter-rater reliabilities for most dimensions have reached substantial to perfect agreement (Chaturvedi & Shweta, 2015; Landis & Koch, 1977). Although some dimensions (i.e., refinement and solution alignment) yielded lower inter-rater reliability coefficients, all discrepancies were reconciled through discussion between the two raters, adhering

closely to the coding scheme to ensure consistency and rigor in the final coding.

#### 4.5. Creativity of problem-solving reports

We employed both computational methods and human ratings to evaluate the creativity of problem-solving reports. The computational methods assessed semantic originality and idea density (Bower et al., 2023). Human raters evaluated creativity in terms of idea novelty and writing novelty.

**Computational Indicators of Creativity.** The computational methods include two approaches (i.e., semantic originality and idea density metrics) with three indicators (i.e., Originality, CPIDR and DEPID) that are commonly used to evaluate creativity in written text (Bower et al., 2023).

**Originality.** Semantic originality was assessed using centroid-based cosine similarity, wherein each problem-solving report embedding was compared to the centroid of all submissions. We implemented this calculation using the *sentence-transformers* Python package, which allows for efficient generation of contextualized sentence embeddings (Reimers & Gurevych, 2019). Then, this method quantifies the uniqueness of a group report's content relative to the collective dataset (all the group reports in our case), with lower similarity scores indicating higher originality. We computed originality by subtracting the cosine similarity score of each proposal from one, such that higher values indicate greater semantic divergence from the central tendency of all proposals, and thus the higher values indicate higher originality. Such semantic measures have been instrumental in capturing the novelty aspect of creativity (Bower et al., 2023).

**CPIDR.** We analyzed idea density using two computational tools available in the Python package *ideadensity* (Robison, 2024): the Computerized Propositional Idea Density Rater (CPIDR) and the Dependency-based Propositional Idea Density (DEPID) method. CPIDR calculates the ratio of propositions to total words, where a proposition refers to an expressed idea or meaning-bearing unit, such as a subject–verb–object clause (e.g., “students use AI”), a predicate (e.g., “should be avoided”), or a logical connector indicating contrast or causality (e.g., “however,” “because”). This indicator can effectively gauge the informational richness of given text (Brown et al., 2008).

**DEPID.** DEPID leverages syntactic dependency parsing to identify unique propositions. For instance, in the sentence “Students should cite AI tools,” distinct propositions might include “students cite tools” and “tools are AI.” In a more complex sentence like “Although AI can assist with research, students must critically evaluate its output,” DEPID may identify several distinct propositions such as “AI assists research,” “students evaluate output,” and “evaluation must be critical.” DEPID captures the underlying semantic structure of ideas, offering a nuanced perspective on the complexity and depth of ideas presented (Sirts et al., 2017). High idea density scores from these tools suggest a sophisticated articulation of concepts, which is indicative of creative problem-solving (Brown et al., 2008; Sirts et al., 2017).

**Human Rating of Novelty.** We also conducted a human rating procedure to assess reports' creativity. Two human raters independently reviewed all reports and first summarized the key ideas proposed in each submission using short keywords (e.g., alternative assessment, AI workshops, penalty system). Drawing on prior work on scoring novelty and originality (e.g., Mozaffari, 2013; Plucker et al., 2004), the raters evaluated creativity along two dimensions that capture both the content and expression of students' ideas (see Table 2 for rubrics). First, two human raters independently assessed idea novelty. Each idea was rated on a 1–2 scale, where 1 indicated that the idea was common across many reports and 2 indicated that the idea was novel within the context of this dataset. Because reports varied in the number of ideas presented, we calculated an average idea novelty score for each report to avoid conflating creativity with simply producing more ideas. Second, writing novelty was also rated on a 1–2 scale, with 1 representing general or shallow explanations, and 2 reflecting elaborated reasoning that demonstrated unique or original explanations. After coding and discussion, the raters reconciled disagreements. We did not compute inter-rater reliability because coders worked collaboratively to reach a consensus score for each report, and the sample size was small. We combined average idea novelty with writing novelty into a single novelty score to capture both the originality of the ideas and the distinctiveness of how they were articulated. This integrated measure offers a more balanced and meaningful representation of creativity than either component alone:

$$\text{Novelty} = (\text{Sum of Idea Novelty} / \text{Number of Ideas}) + \text{Writing Novelty}$$

**Table 2**

Rubric for human rating of novelty.

| Dimensions      | Levels and Descriptions                                                            | Examples                                                                                                                                                                                                                                                                                                                                                              |
|-----------------|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Idea novelty    | 2 – novel ideas are being presented                                                | AI assistant chat; in-class writeup as baseline for comparison                                                                                                                                                                                                                                                                                                        |
|                 | 1 – the ideas are common in the reports                                            | AI literacy workshops; guidance on AI usage; change assessment format                                                                                                                                                                                                                                                                                                 |
| Writing novelty | 2 – the writing is elaborated and demonstrates unique reasoning                    | “All will benefit from an AI assistant chat as instructors can set certain rules on the AI, where they would provide some assistance and only the minimum feedback and answer, they could also have AI ask questions back to the students so they may think about it, the process repeats until the student may get a satisfied understanding to creating a solution” |
|                 | 1 – the writing is general and simple with no elaborated or original explanations. | “Offer optional workshops on AI literacy that teach students how to use AI responsibly for brainstorming or grammar checks rather than content generation”                                                                                                                                                                                                            |

#### 4.6. Adoption of Generative AI's outputs

We assessed the similarity between 29 group reports and the associated chat histories with Generative AI to understand how students incorporated AI-generated content into their final outputs. We computed the cosine similarity between each group's final report and its AI chat transcripts. A higher similarity score suggests closer alignment with the AI-generated outputs, while a lower score implies greater independence. This method provides an objective indicator of students' reliance on Generative AI in collaborative problem-solving contexts.

#### 4.7. Statistical analyses

Following these analyses, we first conducted regressions to explore whether the intervention improved students' perceived critical thinking (RQ1). Then, we conducted more regression analyses to explore the effect of intervention on critical thinking and reliance behaviors, with trust and disciplinary backgrounds as control variables (RQ2). We conducted a *t*-test to compare participants' trust in Generative AI and found that the trust level in the comparison group is significantly higher than that in the experiment group ( $t(225) = 2.61, p < .05$ ). In addition to trust, the participants have unbalanced disciplinary backgrounds, with more students from STEM disciplines than students from non-STEM disciplines. Thus, trust and disciplinary background were included in the analysis as control variables. Further, we conducted correlation analysis among the group-level indicators, including the adoption of Generative AI's outputs as an objective indicator of AI reliance, quality of problem-solving, and creativity (RQ3). Lastly, to explore the effect of intervention on collaboration outcomes, we conducted a series of *t*-tests on indicators associated with collaboration outcomes, including 10 indicators for quality of problem-solving, 5 indicators for creativity, and the adoption rate of Generative AI (RQ4).

**Contextualizing Statistical Analysis.** To complement the statistical findings of group-level analysis and deepen our understanding of human-AI collaboration, we conducted a qualitative analysis of the 38 group reflections submitted alongside the problem-solving reports. Specifically, these reflections captured students' descriptions of how they used Generative AI during the activity and their perceptions of its helpfulness or limitations. This qualitative data deepens our understanding of how Generative AI has been used, as well as whether people have carefully evaluated the output. By complementing these qualitative insights with quantitative results, we aimed to contextualize the observed outcomes and gain a deeper understanding of the underlying collaborative dynamics between students and Generative AI.

### 5. Results

**RQ1** Does the intervention improve students' perceived critical thinking during human-AI problem-solving, considering individual differences in trust and disciplinary background?

We first conducted an intervention check to understand if the intervention, targeting critical thinking, has indeed improved students' perceived critical thinking. The regression model predicting critical thinking, with trust and disciplinary background as controlling variables, was not statistically significant,  $F(3, 222) = 0.53, p = .65$ , suggesting that none of the predictors, including condition, explained meaningful variance in self-reported critical thinking. This result shows that the intervention does not have an effect on students' perceived critical thinking.

**RQ2** How does the intervention impact individual students' reliance behaviors on Generative AI, considering the effect of trust and disciplinary background?

Table 3 presents the descriptive statistics and correlation for individual-level factors. Critical thinking is positively correlated with collaborative, reflective, and cautious use, which is consistent with our prior finding that these types of use of Generative AI might be indicators of desirable reliance behaviors (Hou et al., 2025). Trust is positively associated with thoughtless and reflective use. Reflective and cautious use are strongly correlated ( $r = .66, p < .05$ ), suggesting they may share underlying evaluative processes.

Table 4 presents the regression results examining the associations between conditions (intervention vs. comparison), trust in Generative AI, and disciplinary background with four types of reliance behaviors (i.e., thoughtless, cautious, collaborative, and reflective use). The regression model predicting thoughtless use was statistically significant,  $F(3, 222) = 4.74, p < .05$ , with an adjusted

**Table 3**  
Descriptive statistics and correlations among individual-level factors.

| Variables            | M(SD)      | Skewness | Correlation |      |      |      |      |   |
|----------------------|------------|----------|-------------|------|------|------|------|---|
|                      |            |          | 1           | 2    | 3    | 4    | 5    | 6 |
| 1. Critical Thinking | 5.40(.68)  | -.07     | –           |      |      |      |      |   |
| 2. Trust             | 3.12(.57)  | -1.06    | -.7         | –    |      |      |      |   |
| 3. Collaborative     | 3.45 (.78) | -.81     | .18*        | .10  | –    |      |      |   |
| 4. Thoughtless       | 2.86 (.88) | .17      | -.09        | .20* | .24* | –    |      |   |
| 5. Reflective        | 3.46 (.72) | -.67     | .18*        | .25* | .41* | .39* | –    |   |
| 6. Cautious          | 3.39(.65)  | -.41     | .19*        | .06  | .34* | .16  | .66* | – |

\* $p < .05$ .

**Table 4**

The regression coefficients and statistics with critical thinking and reliance behaviors as dependent variables.

| Predictors                     | Thoughtless use                              |     |        | Cautious use              |     |       | Collaborative use                            |     |        | Reflective use                               |     |        |
|--------------------------------|----------------------------------------------|-----|--------|---------------------------|-----|-------|----------------------------------------------|-----|--------|----------------------------------------------|-----|--------|
|                                | beta                                         | se  | t      | beta                      | se  | t     | beta                                         | se  | t      | beta                                         | se  | t      |
| Trust                          | .31                                          | .11 | 2.85** | .08                       | .08 | -.42  | .13                                          | .09 | 1.32   | .31                                          | .08 | 3.56** |
| Disciplinary Background [STEM] | -.05                                         | .13 | -.44   | .09                       | .10 | .98   | .27                                          | .11 | 2.31** | -.14                                         | .09 | -1.54* |
| Condition [Experiment]         | -.20                                         | .11 | -1.76* | -.03                      | .08 | 1.00  | -.02                                         | .10 | -.21   | .31                                          | .08 | 3.56   |
| Intercept                      | 2.01                                         | .36 | 5.52** | 3.07                      | .27 | 11.06 | 2.84                                         | .32 | 8.65** | 22.38                                        | .29 | 8.10** |
| <b>Model Specifics</b>         | F(3.222) = 4.74, $p < .05$ , Adj $R^2 = .05$ |     |        | F(3.222) = .90, $p = .43$ |     |       | F(3.222) = 2.83, $p < .05$ , Adj $R^2 = .02$ |     |        | F(3.222) = 8.09, $p < .05$ , Adj $R^2 = .09$ |     |        |

Note. \* $p < .1$ ; \*\* $p < .05$ .

RQ3 On the group level, how do human-AI collaborative outcomes, including problem-solving qualities, adoption of Generative AI's outputs, and creativity, correlate with each other?

$R^2$  of .05. Both trust ( $\beta = .31, t = 2.85, p < .01$ ) and condition ( $\beta = -.20, t = -1.76, p = .08$ ) were meaningful predictors. Higher trust in Generative AI was associated with greater thoughtless use, while students in the intervention condition reported marginally lower levels of thoughtless use compared to those in the comparison group. This suggests the intervention reduced thoughtless use of AI, although the effect was only marginally significant. The model for cautious use was not significant,  $F(3, 222) = 0.90, p = .43$ . The model for collaborative use reached significance,  $F(3, 222) = 2.83, p < .05$ , although only disciplinary background emerged as a significant predictor ( $\beta = .27, t = 2.31, p < .05$ ), with STEM students reporting higher collaborative use of Generative AI than non-STEM students. Trust and condition were not significant predictors of collaborative use. Lastly, the model for reflective use was statistically significant,  $F(3, 222) = 8.09, p < .05$ , with an adjusted  $R^2$  of .09. Trust ( $\beta = .31, t = 3.56, p < .01$ ) is a significant positive predictor of reflective use. Disciplinary background ( $\beta = -.14, t = -1.54, p = .05$ ) is a marginally significant negative predictor of reflective use. The results indicate that people with higher trust are more likely to engage in reflective use of Generative AI, and people from non-STEM disciplines are more likely to engage in reflective use of Generative AI than students from STEM disciplines.

Table 5 presents the descriptive statistics and correlations of group-level variables. In total, we analyzed 38 group reports and 28 chat histories (due to partial data submission). Notably, the two indicators of idea density, namely CPIDR and DEPID, were significantly positively correlated ( $r = .42, p < .05$ ), suggesting internal consistency in the measurement of idea densities. Additionally, Generative AI Adoption was significantly negatively correlated with originality ( $r = -.49, p < .05$ ), indicating that group reports that more directly adopted AI-generated content are associated with less originality. These correlations support the interpretive utility of CPIDR, DEPID, originality, and AI adoption as group-level indicators of creativity. Moreover, Generative AI Adoption showed moderate positive correlations with several dimensions of problem-solving quality, including Impact Analysis ( $r = .45, p < .05$ ), Solution Formation ( $r = .41, p < .05$ ), and Refinement ( $r = .49, p < .05$ ). These results suggest that although Generative AI adoption is associated with lower originality, it is related to structured reasoning or elaboration in some subcomponents of problem-solving.

The human-rated novelty was significantly positively correlated with semantic originality ( $r = .36, p < .05$ ) and significantly negatively correlated with Generative AI Adoption ( $r = -.35, p < .05$ ), suggesting that texts judged by humans as more creative were also more semantically divergent and less directly derived from AI outputs. As expected, correlations with idea density metrics were lower and non-significant (CPIDR:  $r = .10, p = .57$ ; DEPID:  $r = .19, p = .25$ ), reflecting that idea density metrics complement novelty assessment by emphasizing informational richness rather than novelty. The findings demonstrate that the computational methods align well with human judgment of creativity but could reflect different dimensions of creativity.

To examine whether the intervention influenced collaborative outcomes, we conducted independent-sample t-tests comparing groups in the intervention and comparison conditions (Table 6). Significant group differences emerged across several creativity-related indicators. The intervention group demonstrated significantly higher originality ( $t(31) = -2.06, p < .05$ ), higher idea density, as measured by DEPID ( $t(31) = -2.16, p < .05$ ) and marginally by CPIDR ( $t(31) = -1.77, p = .08$ ) and higher human-rated novelty ( $t(31) = -2.12, p < .05$ ). Generative AI Adoption was significantly lower in the intervention group ( $t(22) = 2.50, p < .05$ ), indicating that the intervention has successfully reduced passive adoption of AI-generated content. Results showed no significant group differences in most dimensions of problem-solving performance. However, the comparison group scored significantly higher on the Evaluation dimension ( $t(31) = 2.49, p < .05$ ), suggesting that the comparison group provides more elaborate evaluation during problem-solving.

Further, we used students' reflections to supplement the quantitative analyses of the group-level variables. We noticed that the experiment groups used GPT-based chatbots for feedback and improvement and thus had lower Generative AI adoption and higher creativity. For example, one group said, "We have used the GPT to clarify our doubts and reestablish parameters and guidelines set by the question."; another group said, "We used GenAI to brainstorm and further enhance the solutions we have thought of.". While for the comparison groups, students used Generative AI more extensively and showed direct adoption of Generative AI's content. For example, one group said, "Our group uses GenAI to come up with a solution and list out the strengths and weaknesses, as well as possible consequences of the solutions."; another group said, "Generative AI can generate ideas and solutions for the case above. AI helps detect the concerns faced by students, professors, and institutions. We found Generative AI's outputs very helpful as it did problem solving and analysis (Time saving), also it helped us in comprehensive coverage (Identify all the aspects and possibilities), and idea generation (Solution)". Students' reflections are consistent with the findings from the quantitative analyses. The quantitative and qualitative results suggest that the intervention strategies have led students to engage with Generative AI less thoughtlessly and maintain human autonomy and independence, compared to students who used Generative AI freely.

**Table 5**  
Descriptive statistics of group-level variables.

| Variable                   | Correlation |            |      |      |       |       |       |      |      |      |      |      |      |       |      |      |    |
|----------------------------|-------------|------------|------|------|-------|-------|-------|------|------|------|------|------|------|-------|------|------|----|
|                            | n           | Mean (SD)  | 1    | 2    | 3     | 4     | 5     | 6    | 7    | 8    | 9    | 10   | 11   | 12    | 13   | 14   | 15 |
| 1. CPIDR Density           | 38          | .49 (.02)  | –    |      |       |       |       |      |      |      |      |      |      |       |      |      |    |
| 2. DEPID Density           | 38          | .40 (.02)  | .42* | –    |       |       |       |      |      |      |      |      |      |       |      |      |    |
| 3. Generative AI Adoption  | 28          | .58 (.19)  | –.24 | –.27 | –     |       |       |      |      |      |      |      |      |       |      |      |    |
| 4. Originality             | 38          | .04 (.01)  | .13  | .07  | –.49* | –     |       |      |      |      |      |      |      |       |      |      |    |
| 5. Problem Definition      | 38          | 1.55 (.60) | .26  | .12  | .04   | –.35* | –     |      |      |      |      |      |      |       |      |      |    |
| 6. Impact Analysis         | 38          | 1.39 (.75) | .10  | –.18 | .45*  | –.31  | .26   | –    |      |      |      |      |      |       |      |      |    |
| 7. Perspective Integration | 38          | 1.71 (.61) | .21  | .04  | .27   | –.45* | .37*  | .31  | –    |      |      |      |      |       |      |      |    |
| 8. Solution Formation      | 38          | 1.71 (.46) | –.01 | –.08 | .41*  | –.17  | .31   | .30  | .51* | –    |      |      |      |       |      |      |    |
| 9. Evaluation              | 38          | 1.58 (.76) | –.12 | –.17 | .19   | –.31  | –.04  | .28  | .38* | .05  | –    |      |      |       |      |      |    |
| 10. Solution Justification | 38          | 1.37 (.75) | .14  | .11  | .29   | –.05  | .31   | .25  | .33  | .14  | –.02 | –    |      |       |      |      |    |
| 11. Feasibility            | 38          | 1.13 (.91) | .10  | .07  | .14   | –.22  | .30   | .39* | .24  | .22  | .40* | .39* | –    |       |      |      |    |
| 12. Refinement             | 38          | 1.89 (.86) | .03  | .00  | .49*  | –.18  | .17   | .20  | .30  | .29  | .35* | .48* | .34* | –     |      |      |    |
| 13. Conflict Analysis      | 38          | .87 (.91)  | –.03 | –.17 | .18   | –.35* | .05   | .12  | .38* | .10  | .59* | .25  | .26* | .46*  | –    |      |    |
| 14. Measurable Success     | 38          | 1.37 (.75) | –.15 | –.02 | .04   | –.27  | .43*  | .52* | .25  | .09  | .43* | .36* | .54* | .45*  | .31  | –    |    |
| 15. Human-rated Novelty    | 38          | 2.5 (.73)  | .10  | .19  | –.28  | .36*  | –.35* | –.09 | –.12 | –.08 | –.14 | –.21 | .09  | –.40* | –.26 | –.03 | –  |

\* $p < .05$ .



**Table 6**

T-test between the comparison and intervention conditions for all collaborative outcomes.

|                                        | Mean       |            | df | t       |
|----------------------------------------|------------|------------|----|---------|
|                                        | Comparison | Experiment |    |         |
| Total                                  | 23         | 15         | –  | –       |
| Quality of Problem-solving performance |            |            |    |         |
| Problem Definition                     | 1.50       | 1.47       | 31 | .11     |
| Impact Analysis                        | 1.42       | 1.31       | 31 | .40     |
| Perspective Integration                | 1.85       | 1.57       | 31 | 1.25    |
| Solution Formation                     | 1.71       | 1.73       | 31 | –.13    |
| Evaluation                             | 1.92       | 1.31       | 31 | 2.49**  |
| Solution Justification                 | 1.42       | 1.31       | 31 | .42     |
| Feasibility                            | 1.14       | 1.00       | 31 | .42     |
| Refinement                             | 1.85       | 1.89       | 31 | –.11    |
| Conflict Analysis                      | 1.21       | .68        | 31 | 1.69    |
| Measurable Success                     | 1.33       | 1.39       | 31 | –.23    |
| Creativity                             |            |            |    |         |
| CPIIDR Density                         | .47        | .49        | 31 | –1.77*  |
| DEPID Density                          | .38        | .40        | 31 | –2.16** |
| Originality                            | .03        | .04        | 31 | –2.06** |
| Generative AI Adoption                 | .69        | .50        | 22 | 2.50**  |
| Human-rated Novelty                    | 2.19       | 2.70       | 31 | –2.12** |

\* $p < .1$ ; \*\* $p < .05$ .

## 6. Discussion

This study investigated how a classroom-based intervention, designed to promote critical thinking, influenced students' reliance behaviors, quality of problem-solving, and creativity during human-AI collaboration. Drawing on prior literature highlighting effective strategies for promoting critical thinking (e.g., [Abrami et al., 2015](#); [Liu & Pásztor, 2022](#)), the intervention integrated three strategies: authentic instruction, reflective dialogue, and mentoring. Specifically, students were prompted to identify potential biases in AI outputs (authentic instruction), were supported by a GPT-based peer tutor that encouraged reflection and deep inquiry (mentoring) and were instructed to engage in role-play to evaluate solutions from multiple stakeholder perspectives (dialogues). These strategies were embedded within a problem-solving activity, where students collaborated to define problems, explore information, and propose solutions with the support of Generative AI.

We first examined whether the intervention designs improved students' critical thinking and found no significant effect on their self-reported outcomes. A possible reason is that critical thinking is less likely to change through short-term interventions. A meta-analysis of interventions on critical thinking in higher education reveals that short-term interventions (one to four weeks) have no effect on critical thinking ([Liu & Pásztor, 2022](#)). The current study only implemented the intervention for 2 hours in one session. Another explanation concerns the intensity and implementation of the intervention. Although brief, the intervention was cognitively demanding. Students were required to engage in authentic instruction, structured dialogue, and AI-mediated mentoring within a short timeframe. Thus, students might have engaged unevenly with the available critical thinking scaffolds, for instance, some participating more actively in group dialogues, others interacting more frequently with the GPT-based peer tutor. These variations in engagement could have led to different levels of critical thinking in the experiment condition and therefore weaken the overall observed intervention effect ([Horner et al., 2006](#)). Third, the non-significant result may stem from limitations in the measurement approach. The current study relied on a self-report measure adapted from the Motivated Strategies for Learning Questionnaire (MSLQ; [Pintrich et al., 1993](#)). Because problem-based learning inherently fosters critical thinking ([Anggraeni et al., 2023](#)), students in both conditions may have perceived themselves as engaging in critical thinking regardless of the intervention conditions. Consequently, the subtle differences in critical thinking between conditions might not be fully captured through a self-reported survey, but may be better observed in students' behaviors, such as reflective or thoughtless use of AI. It is therefore plausible that students acted more reflectively without registering an increase in perceived critical thinking.

At the individual level, the intervention was associated with a marginal reduction in thoughtless use of Generative AI. The finding suggests that at the individual level, participants themselves perceived less thoughtless use of Generative AI, including passive adoption of AI-generated content. Such a self-perceived difference in the thoughtless use of Generative AI is also observed through a group-level analysis of students' group reports and chat histories. We found that the experiment groups have less direct adoption of Generative AI-generated content. These results suggest that intervention strategies can shift students' reliance patterns away from passive adoption. The results highlight the potential of instructional strategies to change reliance behaviors on Generative AI, aligning with the studies that emphasize careful pedagogical designs when integrating Generative AI in the classroom ([Moussa, 2024](#); [Zhai et al., 2024](#)). Additionally, we found that trust in AI is a positive predictor for both reflective and thoughtless use of Generative AI, which aligns with our previous finding ([Hou et al., 2025](#)). In that earlier study, we observed that trust can lead to increased overall usage of Generative AI, but the type of usage (i.e., reflective or thoughtless), is related to students' critical thinking. Therefore, it is important to promote critical thinking to channel trust toward constructive rather than thoughtless use of Generative AI.

At the group level, this study investigated the effect of intervention on collaborative outcomes, including problem-solving quality

and creativity. The correlations between group-level variables show that direct adoption of Generative AI's content is positively associated with some aspects of problem-solving quality, including Impact Analysis, Refinement, and Solution Formation. Also, groups in the comparison condition, which demonstrated more thoughtless use of Generative AI, scored higher on certain aspects of problem-solving quality, particularly Evaluation. These results suggest that Generative AI use may support analytical clarity and coherence. In students' reflections, some also stated that "Generative AI provided us with better flow, comprehension, description of our solutions." The findings reflect previous studies which show that AI-based chatbots can improve the organization, sufficiency, and elaboration of students' arguments in debates (Guo et al., 2023). One study found that students engaged in more knowledge-based dialogue and elaborated discussions with chatbots, whereas they engaged in more social dialogues in peer interactions (Song et al., 2025). Engaging more in knowledge-based dialogues with Generative AI might explain why the comparison group can achieve better in elaboration during problem-solving. In contrast, the experiment groups, which were instructed to engage more with their peers to maintain human autonomy, might allocate more time to social regulation but less time for elaborating and analyzing their solutions to problem-solving. Another explanation is, as Zhai et al. (2024) observed, Generative AI exhibits resilience against cognitive intensity, enabling it to perform consistently across tasks. Human learners, in contrast, may face cognitive overload when simultaneously engaging in reflection, integration, and articulation during problem-solving.

Aside from the quality of problem-solving, originality is another critical factor of human-AI problem-solving. The moderate correlation between the human novelty score and the semantic distance indicates that both approaches capture some shared aspects of creativity. Human ratings emphasize contextual judgment and the perceived uniqueness of ideas, whereas computational methods quantify semantic distance at scale. Both human ratings and semantic distance could measure closeness among concepts and ideas. Such relatedness between ideas could be apparent to human judgment and can also be assessed through computational methods (Runco et al., 2025). The weaker associations between idea density measures and human novelty scores indicate that computational metrics can also assess dimensions of creativity, such as information richness, that may be difficult for humans to capture. Computational methods can evaluate a large amount of information to quantify the idea richness (Brown et al., 2008). Together, these findings suggest that computational metrics can complement human assessments and potentially provide a practical alternative for evaluating creativity in large-scale text analyses.

Groups in the intervention condition, who had lower thoughtless use of Generative AI, demonstrated significantly higher levels of creativity as measured by originality and idea density. Adoption of Generative AI's content is also negatively associated with originality. These findings contribute to the ongoing research around the role of Generative AI in creativity. While some studies report that AI assistance may inhibit creative outputs (Baltà Salvador, El Madafri, Braso Vives, & Peña Carrera, 2024), others highlight its potential to generate novel or unconventional ideas. For instance, Fu et al. (2024) demonstrated that GPT-produced visual designs that are more divergent and surprising, although not necessarily more useful. Similarly, Urban et al. (2024) found that students who used GPT generated more creative problem-solving solutions than those who did not, as measured by higher elaboration (amount of detail), originality (novelty of ideas), and quality (alignment with task goals). While the current study also observed that Generative AI use improved the quality of problem-solving, we found that originality was *lower* among groups that relied more heavily on Generative AI. This discrepancy may stem from the difference in measurement and experiment setups. Urban and colleagues (2024) relied on expert ratings to evaluate creativity in a manufacturing product improvement task, whereas our study employed computational measures such as idea density and semantic uniqueness in policy proposals. Another explanation is that students in our experiment groups were encouraged to interact *critically* with AI, discouraging passive acceptance of AI outputs and leading to higher originality. These findings build upon Medeiros et al. (2025), who cautioned that low-quality or misaligned AI assistance can hinder human creativity in problem-solving.

Therefore, our findings suggest that the benefits of AI on creativity are conditional, depending on *how* people use Generative AI. In our context, the problem-solving task did not explicitly require students to generate innovative solutions. Yet, even within such boundaries, students who engaged more critically and independently produced more creative outputs in terms of originality and idea diversity. This may be attributed to the relatively predictable nature of GPT responses compared to the associative leaps enabled by human cognition (Beatty & Kenett, 2023). When students are prompted to contextualize and revise AI output, rather than adopt it uncritically, they may contribute more distinct and meaningful ideas. Thus, while general Generative AI use can enhance elaboration and clarity, it is thoughtful use that appears to foster creativity.

## 7. Implications

Our findings have important implications for ongoing research and educational practice concerning human-Generative AI collaborations. For research, our study is among the first to intervene in students' reliance behaviors through instructional designs. Our findings contribute to the ongoing research on the effect of Generative AI on learning outcomes, such as problem-solving qualities and creativity. Our findings suggest that Generative AI may be highly capable of producing detailed and goal-aligned responses, thus elevating quality indicators (Urban et al., 2024). However, human inputs remain crucial for generating novel and contextually grounded ideas. Thus, effortful engagement with Generative AI is vital to unlocking creative potential in human-AI problem-solving. The finding also reinforces Bearman, Ryan, and Ajjawi's (2023) notion of "altering authority," where AI's expanding role in intellectual tasks reshapes the purpose and value of student contributions. In this view, the emphasis is on how people engage with Generative AI, instead of whether it should be used.

For practitioners, the study affirms the importance of designing interventions that scaffold critical thinking when using Generative AI. The evidence suggests that even short interventions can reduce passive overreliance and stimulate creative outputs. To translate such interventions into classroom practice, instructors can guide students to engage critically by questioning, refining, and contextualizing AI-generated ideas. At the same time, learning activities should frame AI as a collaborative partner in open-ended problems that invite exploration and judgment, where students contribute insights that go beyond what AI can offer. Assessment should also evolve to value how students reason with and about AI, emphasizing justification, integration of perspectives, and creative synthesis rather than the accuracy or polish of AI-assisted outputs. At the institutional level, the findings advocate for professional development initiatives that equip educators to integrate AI thoughtfully and ethically in their teaching and enable students with time and space to think critically and creatively in their learning tasks, even when Generative AI is pervasive.

## 8. Limitations and future directions

This study has several limitations. First, the intervention was implemented during a single 2-h session, which may have been insufficient to produce measurable changes in students' self-reported critical thinking. Future studies should therefore consider repeated sessions or embedding the strategies across an entire course to examine how critical thinking skills can be cultivated and reinforced in sustained ways. Longitudinal interventions are more likely to yield measurable changes in critical thinking. Additionally, tracking changes over time could provide insight into whether and how reliance patterns evolve over time, and how students adapt their strategies to use AI through repeated exposure. Furthermore, assessing critical thinking through self-report alone may not capture the behavioral and process-oriented aspects of how students reason with AI. Future studies should incorporate process-tracing methods such as log data or think-aloud to more comprehensively capture critical thinking. This would yield richer insight into how students evaluate and regulate AI contributions during problem-solving.

Second, while the problem-solving task required students to develop policy solutions, it did not explicitly prompt them to generate creative or novel ideas. Nevertheless, we observed that passive reliance on Generative AI was associated with more predictable and less original responses, suggesting that thoughtless use may suppress creativity when not explicitly required. Future research should incorporate problem tasks that explicitly require creative ideation or divergent thinking, such as open-ended design challenges or innovation-focused scenarios. This design would help further uncover how reliance behaviors interact with creativity in human-Generative AI collaborative learning.

In addition, the current study relied solely on group reports to assess collaborative outcomes. We apply automated text-mining techniques to approximate creativity and human labeling of problem-solving quality. Although such measures offer scalability and objectivity, they may not fully capture the contextual and domain-sensitive aspects of problem-solving. For creativity, triangulating computational indicators with rubric-based or expert ratings in future research would provide a more robust and nuanced evaluation of creativity outcomes. While assessing problem-solving quality with a coding scheme is common, it provides only one perspective on collaborative outcomes. Future studies should incorporate multimodal evidence, such as process data, recorded discussions, or observational measures, to provide a more comprehensive evaluation of problem-solving quality.

Finally, the sample for group-level analysis is relatively small, with only 38 groups in total, and the individual participants consisted of relatively homogeneous university students from a single institution, which may limit the generalizability of the findings to more diverse educational or cultural settings. The relatively small sample size also limits our ability to do multilevel data analysis on students' individual factors and collaborative performance. In addition, because the intervention combined three strategies simultaneously, we were not able to isolate the differential contributions of authentic instruction, structured dialogue, and mentoring. Although this integrated design was grounded in prior meta-analytic evidence showing that combined approaches yield the strongest effects (Abrami et al., 2015), future studies with larger and more diverse samples could vary these strategies or track their use over time to identify which elements are most influential so as to improve the practical utility of research results. Future studies should implement the intervention across diverse educational contexts, including different disciplines, institutions, and cultural settings, with a larger sample size to explore multilevel effects and examine whether the AI reliance typology generalizes beyond this setting.

## 9. Conclusion

This study advances our understanding of how instructional interventions on critical thinking, including authentic instruction, GPT-based peer tutors, and role-playing, can shape students' reliance behaviors and collaborative outcomes during human-AI problem-solving. Although the short intervention did not significantly enhance critical thinking, it effectively reduced thoughtless reliance and direct adoption of AI-generated content. Notably, the intervention fostered greater creativity in students' solutions, underscoring the value of promoting effortful engagement with Generative AI. These findings highlight that how students use AI, rather than whether they use it, is critical to supporting thoughtful learning and creative outcomes in PBL environments. Future research should further explore scalable, longer interventions targeting students' critical thinking across diverse contexts and learning tasks to cultivate sustained critical engagement with AI and examine its impacts.

## CRediT authorship contribution statement

**Chenyu Hou:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Gaoxia Zhu:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Yanzhi Liu:** Writing – review & editing, Formal analysis. **Vidya Sudarshan:** Writing – review & editing, Investigation, Funding acquisition. **Josephine Leng Leng Chong:** Writing – review & editing, Investigation. **Fannie Yifan Zhang:** Writing – review & editing, Investigation. **Michael Yong Heng Tan:** Writing – review & editing, Investigation. **Yew Soon Ong:** Writing – review & editing, Resources, Investigation, Funding acquisition.

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## Declaration of competing interest

The authors declare no potential conflict of interest in the work.

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## Appendix A

### *Problem-Solving Activity – Practice guide for teaching and learning with Generative AI*

NTU has implemented guidelines on using AI tools on campus, emphasizing the development of well-rounded students capable of critical thinking, creativity, and leadership in environments enriched by AI technologies. The objectives of CC0002 – Navigating the Digital World course align with these guidelines by integrating Generative AI into learning activities as a form of “[assessment as learning](#)”. However, instructors have observed issues concerning students’ AI usage.

To address the issues, the ICC office is now compiling a practice guide for teaching and learning with Generative AI. The office is brainstorming and collecting solutions to address Generative AI issues that occurred during teaching and learning in ICC courses. You and your teammate are invited to **provide specific solution(s) to help address the cases of using Generative AI in the ICC courses**. You can focus on **one** of these cases and work out a solution. If your solution is well-rounded, we will include it in the practice guide that will be distributed to instructors in the next semester. Thank you for your contribution!

#### • Case #1

An instructor of CC0001 mentioned that students excessively use AI in essay writing. Instead of writing the essay independently, some students input the main ideas and examples in bullet points, along with the assignment criteria, and then use the AI to generate content directly. The instructor had told the class that this graded assignment would be checked by AI detection tools. When grading the essays, the instructor found a handful of cases where the percentages in the AI detection reports were concerningly high. What should the stakeholders involved do in this case?

#### • Case #2

A student who took the CC0002 course last semester reported a concern regarding the group project. She found that one of her teammates used AI-generated content for his part of the assignment, which would harm the integrity of the whole assignment. Despite her initial request for the teammate to refine his part, the teammate persisted in using AI to finish to group assignment. This student reported the situation to the instructor and sought intervention. Some other students expressed similar concerns about whether they will be penalized for their team members’ inappropriate use of AI in the group assignments. What should the stakeholders involved do in this case?

## Appendix B

The coding scheme for assessing problem-solving performance.

| Coding Category                     | Description                                                                                                                                                                                                                                                  | Examples                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Stage 1: Problem Description</b> |                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Problem Definition                  | 2 – Clearly define the problem and identify the negative consequences.<br>1 – the problem is only briefly mentioned with no or limited explanations on the consequences.<br>0 – Not answered.                                                                | 2 – “The main issue here is the inappropriate use of AI tools by students to complete a graded assignment, potentially compromising academic integrity and the authenticity of student work. Students input minimal information, allowing AI to generate content that they submit as their own. This raises questions about the fairness and effectiveness of assessments within the course.”<br>1 – “The core issue is that the student is too dependant on AI to complete his assignment, without thinking of solving anything himself.”                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Impact Analysis                     | 2 – The answer demonstrates the causal reasoning on how it impacts learning.<br>1 – The explanation is brief and no causal reasoning is identified.<br>0 – Not answered.                                                                                     | 2 – “It will affect their overall learning experiences especially when preparing for the working world. Their critical-thinking and problem solving skills will greatly deteriorate, and during work when problems arise, they would be depending on others to provide a solution as they could not think of one themselves.”<br>1 – “When students use Generative AI unethically, it infringes on the purpose of an undergraduate education - which is to obtain knowledge and learn. Using Generative AI does not reinforce active learning into the students, thus lessening the learning effectiveness of the subject matter.”                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Perspective-Taking                  | 2 - Students list different perspectives of students, ICC administrators and instructors,<br>1 – Students only mention the perspective of 1 or 2 roles<br>0 - No mention of different perspectives.                                                          | 2 – “Students key concerns: If many students depend on AI to provide solutions or ideas, everyone may have the same idea without thinking of any.<br>Instructors: The instructors would have to penalize the student who is using AI to for his assignments as there is no learning to be done when using AI to solve everything.<br>ICC Administrator: They would have to discuss with instructors on ways to prevent the same outcome from happening in the future. If nothing is done, other students would follow suite to use AI for their projects.”<br>1 – “The key concern for students is the lack of integrity, useful outcomes and lessons that they may gain from the course. The key concern for the teachers is wasting resources and energy on helping students who do not want to learn and improve themselves through the course, along with the plagiarism of content that students may not realise and understand.”                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Stage 2: Problem Exploration</b> |                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Solution Alignment                  | 2: The solution is well-structured, clearly addresses stakeholders' concerns, and includes a concrete, actionable plan.<br>1: The solution aligns with stakeholders' concern, but not specific in terms of the action.<br>0: No clear solution is presented. | 2 – “The solution should avoid students from facing a harsh punishment such as failing the module or being graded zero mark since it can negatively affect the students' academic journey and, possibly, their mental health. Besides, the solution also need to provide another way for the instructor to accurately evaluate those students. Therefore, in our opinion, one of the suitable solution is:<br>+ First, give those students a warn on using AI to directly generate their writing, and explain to them again that any action like this will be indicated by AI detection tools. If the students commit this action again, they will be given a heavier punishment such as failing them. This will avoid students from severe punishment<br>+ Next, give those students another assignment with similar grading criteria but different content. This make sure that the instructor still can evaluate the students themselves. However, due to their violation, a certain amount of mark should be reduced.<br>+ The students also need to be aware that any action of using generative AI will be detected, and they should only use that for brainstorming or checking grammar.”<br>1 – “Our solution is to educate students on AI tools. By teaching them about their limitations, students will have a greater understanding of AI and its weaknesses. This will allow them to use these tools more responsibly and effectively.” |
| Evaluation                          | 2 - Students discuss both advantages and limitations of their proposal.<br>1 – Students discuss strength or weakness of their proposal.<br>0 – Not answered                                                                                                  | 2 – “The strength is that it encourages students to have ethical responsibility for their actions. The weakness is that while students may be educated on the reasons why they should not heavily rely on AI tools to complete their work, students may resist and still continue using because it is very hard to change mindsets and habits as AI tools boost productivity.”<br>1 – “Long term solution includes the introduction of mandatory AI workshops to ensure students know early on what kind of expectations and restrictions regarding Generative AI is to be followed when doing their assignments. This will not only make students understand what they can or                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

(continued on next page)

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| Coding Category                | Description                                                                                                                                                                                                                                    | Examples                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Conflict Analysis              | 2 – Recognize stakeholders' conflicts and explain the reasons behind<br>1 – Recognize stakeholders' conflicts only<br>0 – Not answered                                                                                                         | cannot do, but it also justifies the strict punishment the instructors and admins may carry out if such a case were to occur in the future.”<br>2 – “Conflict:<br>Instructor vs. Institution: Instructors might want stricter measures to maintain academic integrity, while institutions may prefer educational approaches to avoid excessive penalization.<br>Institution vs. Students: While the institution aims to uphold academic standards, students might feel these policies restrict their autonomy in using emerging tools like AI. Clear guidance can mitigate this conflict.<br>Students vs. Instructors: Some students may feel the approach is overly punitive if their intentions weren't to cheat, while instructors need to maintain fairness for all students.”<br>1 – “Students might want to complete the task in the most efficient manner and might ignore the weakness of AI tools.”                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Stage 3: Solution Presentation |                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Solution Justification         | 2: The justification directly addresses the problem with strong reasoning and clear connections to the identified issue.<br>1: The justification is provided but lacks depth or clear logical reasoning.<br>0: Not answered.                   | 2 – “Students: having students disclose AI usage and reflect on their contributions helps ensure honesty and prevents excessive reliance on AI, thereby encouraging transparency and academic honesty.<br>● Instructors: can better assess individual contributions to ensure fair grading that doesn't penalize innocent team members for the actions of one person.<br>● ICC administrators: setting clear expectations, providing educational resources, and offering reporting mechanisms helps in preventing misuse of AI and preventing future problems.”<br>1 – “Addressing the Core Issue: The solution tackles the academic integrity issue by proposing a warning system that can discourage the misuse of AI. However, it doesn't fully reinforce the importance of independent writing and critical thinking.”<br>2 – “Potential Positive:<br>When students are more educated on responsible AI use, they are able to learn and use the valuable skills in using technology ethically. Furthermore, having a clearer policy with the AI detection tools, it can help instructors fairly assess each student's individual work. A standardized guideline will ensure that both students and instructors in the course have the same understanding of AI being used in assignments.<br>Negative Outcomes:<br>It is possible that students may be scared of being unfairly penalized, especially those who may unintentionally use AI but did not know that it was problematic. Furthermore, instructors may place too much trust in AI detection reports, potentially leading to false positives, making it unfair to students who were genuinely trying their best. By implementing and monitoring these policies, it may place extra workload on administrators.”<br>1 – “Workshops create a support structure for ethical AI use and equip students and instructors with a deeper understanding of AI's role in academia. Adapting assessments to make them more challenging to complete using AI alone helps reduce AI reliance and encourages deeper engagement.<br>But Developing and implementing workshops, resources, and new assessments will require time and budgetary resources that may stretch administrative capacity. And Students who feel competent with AI might feel that the limitations and workshops are unnecessary, especially if they perceive them as “policing” their learning approach.” |
| Refinement                     | 1 point – Students demonstrate the refinement of the proposed solution comparing to the “evaluation” part in previous step<br>1 point – students explain at least one pros<br>1 point – students explain at least one cons<br>0 - Not answered | 2 – “Potential Unintended Consequences<br>● Culture of Distrust: Strict monitoring could create a culture of distrust, impacting the student-instructor relationship.<br>● Possible Shift in Misuse: Students may find ways around detection tools, pushing misuse to areas that are harder to monitor, such as group work or open-book exams.”<br>1 – “However The emphasis on limiting AI could make students feel that using AI is inherently negative, even when it can be a valuable educational tool.”                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Feasibility                    | 2 – Students explain WHY the unintended consequences might happen<br>1 – Students list the unintended consequence<br>0 – Not answered                                                                                                          | 2 – “We can Monitor Academic Integrity cases to see the succes:Observe if there is a reduction in academic integrity issues related to AI, indicating that guidelines and workshops are effective.”<br>1- “To know how successful is the solution is to have the students test out the program and have feedback on whether the AI assistance is useful enough to provide an overall solution.”                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Measurable success             | 2 – Students provide clear measure for success<br>1 – students' measure is not very actionable<br>0 – Not answered                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |



## Data availability

Data will be made available on request.

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